Two-channel duality, fold memory, and arithmetic blindness in polymer contact networks

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(Dated: October 2, 2026)

The contact network of a collapsed polymer is a finite graph, and graph invariants of the kind used in algorithmic class field theory — the Laplacian, its sandpile group, the discrete Hodge decomposition, the Albanese metric of a periodic cover, the Perron branch of the adjacency — can be read off it exactly. We apply them to 82 conformations of the prime-driven topological polymer (PDTP), whose bending stiffness follows the prime gaps, against surrogates that shuffle the gaps (S1) or keep only their smooth prime-number-theorem envelope (S6″). Unweighted invariants, including the exact sandpile group, do not distinguish the arms. The sequence enters through a divisor, the rank of the stiffness on the vertices, and the two channels then separate. In the bulk (Laplacian) channel the divisor energy $D^\top \Delta^+ D$ separates the true chain from its shuffles at $Z_{\rm seq} = +11.4$ in both partially collapsed and fully relaxed globules, while the boundary (Perron) channel separates no pair of arms. The bulk channel reads the envelope: the PNT-only surrogate matches the true chain ($p = 0.76$), and once the smooth positional modes are removed every arm contrast vanishes ($p_{\rm Holm} = 1$ for all 16 envelope-removed tests). A marginal true-versus-S6″ difference seen in an eight-globule pilot ($p = 0.043$) does not replicate on the full library. What survives in every channel and every arm is a strong memory of the globule's own sequence, $z \approx 6$ to 24 against gap shuffles on the same conformation, equal across arms. The folded network remembers its sequence, and the arithmetic it can tell apart is the envelope's. Two textbook identities hold exactly on all 82 graphs: the backbone winding has Albanese energy N, and the spanning-tree count times the closure-flux energy is an integer.

I. INTRODUCTION

A self-avoiding chain whose bending stiffness follows the gaps between consecutive primes (the prime-driven topological polymer, PDTP) collapses into a globule that is measurably less radially sorted than one built from a shuffle of the same gaps [1]. The imprint survives complete relaxation; its correlation length is a property of the readout; and it is carried by the smooth prime-number-theorem (PNT) envelope at the small-prime end of the chain [1, 2]. A companion study showed that no ordering, positional or scattering readout of the chain registers congruence structure beyond sieve composition [3].

Every one of those readouts treats the conformation as a set of positions. A collapsed chain is also a network: beads are vertices, bonds and contacts are edges. Contact maps are the primary data of chromosome conformation capture [4], and the mathematics of finite graphs offers invariants that see more than positions. The algorithmic class field theory of Ref. [5] organizes them into two channels. The *bulk* channel is the Laplacian Δ, whose cokernel has the sandpile (critical) group as torsion [6] and which, on arithmetic quotients of the Bruhat–Tits tree, carries Frobenius traces. The *boundary* channel is the non-backtracking or adjacency operator, whose equilibrium states see only the Perron branch of the spectrum. The first principle of Ref. [5] is that the two channels do not see the same thing: the boundary is arithmetically blind, the bulk is not.

Here we ask what the two channels of the PDTP contact network see of the prime sequence. A contact network is not an arithmetic quotient, so the trace formulas of Ref. [5] do not apply, and we borrow its constructions, not its theorems. The answer has three parts. Unweighted invariants see nothing of the sequence. Weighted by the sequence, the bulk channel sees the PNT envelope and the boundary channel sees no arm difference. And beyond the envelope every channel shows a memory of the globule's own sequence that is the same for primes, shuffles and PNT-only surrogates.

II. THE CONTACT NETWORK AND ITS INVARIANTS

Graph. For a conformation $\{\mathbf{r}_i\}_{i=0}^{N-1}$ the contact graph Y has the backbone bonds $(i, i+1)$ and the non-bonded contacts $|\mathbf{r}_i - \mathbf{r}_j| < 1.5\sigma$, $|i-j| > 1$. Edges are oriented $i < j$; d is the $N \times m$ incidence matrix with $d_{ie} = -1$, $d_{je} = +1$, so that $(d^\top f)_e = f_j - f_i$ and $\Delta = dd^\top$ is the graph Laplacian. The backbone makes Y connected.

Bulk channel. The sandpile group is $\mathrm{Jac}(Y) = \mathrm{Tor}\,\mathrm{coker}\,\Delta$, of order the number $\kappa(Y)$ of spanning trees, with structure read off the Smith normal form [5, 6]. A sequence enters as a degree-zero divisor on the vertices, $D = \mathrm{rank}(k) - \overline{\mathrm{rank}(k)}$, with k_i the bead stiffness. Its class in Jac has an order, and its energy in the Laplacian metric is

$$E(D) = D^\top \Delta^+ D. \tag{1}$$

Envelope removal. Let $P_{\le K}$ project onto the Legendre polynomials of degree $\le K$ in the backbone coordinate

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$x_i = 2i/(N-1) - 1$, orthonormalized on the grid. The envelope-removed divisor is $D_K = D - P_{\le K} D$, and $E_K = E(D_K)$.

Hodge decomposition. A 1-form f on the edges splits orthogonally as $f = d^\top \phi + h$ with $dh = 0$: a gradient plus a cycle (harmonic) part [5, 7]. The coboundary $d^\top k$ of any vertex function is a gradient, so its cycle part vanishes identically on every graph. To obtain a sequence-sensitive cycle energy we use the stiffness-rank difference carried by the contacts only, $f_e = \mathrm{rank}(k)_j - \mathrm{rank}(k)_i$ on contacts and $f_e = 0$ on the backbone, and report the cycle share $H_c = \|h\|^2/\|f\|^2$. Setting f to zero on the backbone breaks the exactness of the full coboundary: f is no longer the gradient of any vertex function, and its harmonic part is the projection of the sequence's stiffness contrast onto the cycle space. Every cycle of Y contains at least one contact, since the backbone alone is a tree, so this part measures how the stiffness contrast circulates around the loops closed by long-range contacts.

Boundary channel. We use the Perron vector v_P of the adjacency matrix and the rank correlation $\rho_P = \mathrm{Spearman}(k, v_P)$.

Albanese energy. Closing the chain by one edge $e^* = (0, N-1)$ gives a unit cell Y^* for a $\mathbb{Z}$-periodic lattice. For a winding cochain α the inverse effective mass of the band bottom is $M^{-1} = (2t/n)\,\alpha^\top \Pi \alpha$, with $\Pi = I - d^{*\top}\Delta^{*+} d^*$ the projector onto the cycle space [5, Ch. 22].

Proposition 1. *(i) For the backbone winding, α = the indicator of the ring, $\alpha^\top \Pi \alpha = N$ for every conformation. (ii) For the closure flux, α = the indicator of e^*, $\alpha^\top \Pi \alpha = 1/(1+R)$, with R the effective resistance between the two chain ends in the open graph Y. (iii) $\kappa(Y^*)/(1+R) = \kappa(Y)$ is an integer.*

Proof. (i) The ring is a cycle, so $\Pi\alpha = \alpha$ and the energy is its length [5, Cor. 22.4.3]. (ii) $\alpha^\top \Pi \alpha = 1 - R^{Y^*}(e^*)$ [5, Thm. 22.4.1], and e^* in parallel with Y gives $R^{Y^*}(e^*) = R/(1+R)$ [8]. (iii) $1 - R^{Y^*}(e^*) = \kappa(Y^* - e^*)/\kappa(Y^*)$ by Kirchhoff's theorem, and $Y^* - e^* = Y$. □

The backbone winding therefore carries no information about the conformation, and the closure flux reduces to the end-to-end resistance of the contact network. Written for the inverse effective mass, (iii) reads $\kappa(Y^*)\,(n/2t)\,M^{-1} = \kappa(Y)$. Multiplying the closed network's inverse effective mass, in its natural unit, by the spanning-tree count of the closed network returns exactly the spanning-tree count of the open one. The band curvature is therefore quantized in units of $2t/[n\,\kappa(Y^*)]$, the discrete-topological quantization of Ref. [5, Cor. 22.4.2]. It is an identity of Kirchhoff type, valid for every graph, and the exact checks of Sec. IV test the computation, not the physics.

III. DATA AND STATISTICS

We use the 82 stored conformations of PDTP v4.0 [1] at $N = 4 \times 10^4$. The *production* globules are partially collapsed coils after an 8000-step anneal: 24 of the true chain, 8 S1 shuffles and 40 S6″ surrogates (exact gap multiset arranged as a PNT scale family, with no prime-specific fluctuation). The *relaxed* globules are melt-density globules after a 10^6-step isothermal hold [2]: 5 true, 5 S1. The production graphs have 7.6–7.9×10^4 contacts; the relaxed graphs 1.88×10^5. All 82 conformations are used in every analysis below.

Own-sequence null. For every globule and every sequence-weighted statistic we compute the same statistic for 20 random permutations of the globule's own gaps, placed on the same conformation, and report $z = (x - \bar{x}_{\mathrm{null}})/s_{\mathrm{null}}$. A large z means that the conformation is adapted to its own sequence.

Arm contrasts. The true arm is one fixed sequence; each surrogate globule has its own sequence. Following PDTP v4.0 [1] we use a sequence-level statistic, $Z_{\mathrm{seq}} = (\bar{x}_T - \bar{x}_S)/[\hat{\sigma}^2_{\mathrm{seq}} + s^2_T/n_T + s^2_S/n_S]^{1/2}$, with the thermal variance estimated from the true arm and $\hat{\sigma}^2_{\mathrm{seq}} = \max(0, s^2_S - s^2_T)$, referred to t_{n_S-1}. Holm [9] and Benjamini–Hochberg [10] corrections are applied within each family of tests.

IV. RESULTS

A. Unweighted invariants are blind

We computed eight invariants that do not use the sequence directly on all 82 graphs: the mean degree (5.8–6.0 in production, 11.4 relaxed), $\ln \kappa$ (5.5–5.6×10^4 and 9.22×10^4), the algebraic connectivity, the adjacency spectral radius, and, on 400-vertex coarse-grained graphs (blobs of 100 beads), the exact sandpile group computed with PARI/GP: its order (10^{363}–10^{374} in production, 10^{558}–10^{563} relaxed), its 2- and 3-ranks, and the order of the sequence divisor in it. Of the 32 sequence-level contrasts (eight invariants, four pairs of arms) the smallest uncorrected p is 0.032, and every Holm-adjusted p equals 1. The divisor's order is within a factor $10^{0.2}$–$10^{0.5}$ of the group exponent in every arm, the same as for the globule's own gap shuffles ($10^{0.1}$–$10^{0.6}$). The arithmetic of $\mathrm{Jac}(Y)$ does not record the sequence.

B. The bulk channel reads the envelope

The divisor energy, Eq. (1), is the first sequence-sensitive invariant (Fig. 1, Table I). In the production globules the true chain lies far above its own shuffles ($z = +33.5$) and so does S6″ ($+28.9$), while S1 barely does ($+2.6$). At the sequence level the true chain separates from S1 ($Z_{\mathrm{seq}} = +11.4$) and from nothing else: it matches S6″ ($Z_{\mathrm{seq}} = +0.31$, $p = 0.76$). The relaxed globules repeat the true-versus-S1 separation ($Z_{\mathrm{seq}} = +11.4$). The channel reads the composition profile along the chain, the PNT envelope, which S6″ shares and S1 destroys.

TABLE I. Sequence-level contrasts of the divisor-energy z (Fig. 1). Full: $K = 0$. Envelope-removed: the four tests $K \in \{4, 8, 16, 32\}$; range of p_{seq}, and Holm-adjusted p over all 16 envelope-removed tests.

		full		envelope removed	
protocol	contrast	Z_{seq}	p_{seq}	p_{seq}	p_{Holm}
production	true − S1	+11.35	9×10^{-6}	0.63–0.94	1
production	true − S6″	+0.31	0.76	0.67–0.81	1
production	S6″ − S1	+8.35	7×10^{-5}	0.38–0.82	1
relaxed	true − S1	+11.40	3×10^{-4}	0.16–0.65	1

The mechanism is quantitative. The Legendre modes up to degree 4 carry 2.9% of $\|D\|^2$ for the true chain, 2.5% for S6″ and 0.2% for S1. Because Δ^+ weights each mode by the inverse of its eigenvalue, and the backbone makes chain-scale profiles the softest modes of the network, this small fraction dominates the energy.

C. Nothing survives envelope removal

Removing the Legendre modes up to degree $K = 4$, 8, 16 or 32 removes every arm contrast (Table I). None of the 16 envelope-removed tests reaches $p_{\mathrm{seq}} < 0.15$, and after correction all have $p_{\mathrm{Holm}} = 1$ and $p_{\mathrm{BH}} = 0.94$. In particular, an eight-globule pilot had shown the true chain above S6″ at $K = 16$ ($p = 0.043$ uncorrected, Welch). On the full library of 24 true and 40 S6″ globules the difference has the opposite sign ($z = +14.3$ against $+15.8$) and $p_{\mathrm{seq}} = 0.81$. The pilot was a fluctuation. The large between-sequence spread of the S6″ class ($\hat{\sigma}_{\mathrm{seq}} \approx 6$ in z) shows why: a handful of surrogate sequences cannot fix the class mean.

D. The boundary channel separates no arms

The Perron correlation ρ_P is also far from its own-sequence null, with the opposite sign ($z = -3.5$, -3.2, -5.0 for true, S1 and S6″ in production; -10.4 and -12.9 relaxed), but it does not distinguish the arms: no contrast survives correction (smallest $p_{\mathrm{Holm}} = 0.16$). The two-channel dichotomy of Ref. [5] therefore has a polymer counterpart: the bulk sees the envelope and the boundary does not. Neither sees anything beyond the envelope.

E. Every channel remembers the globule's own sequence

Figure 2 collects the own-sequence z of four statistics. All are far outside the null band in every arm. The envelope-removed divisor energy has $z \approx 14$ to 21; the contact cycle share H_c, whose values are 0.053–0.055 in production and 0.035 relaxed, has $z \approx 6$ in production and 8–11 relaxed; the Perron correlation has $z \approx -3$ to -13. None separates the arms after correction (cycle share: smallest $p_{\mathrm{Holm}} = 0.56$, relaxed true versus S1).

The conformation is thus strongly adapted to its own stiffness sequence: stiff and flexible segments find contacts that a random rearrangement of the same gaps would not give them. That adaptation is the generic physics of a semiflexible collapse. A shuffled sequence folded by the same dynamics is adapted to itself just as strongly. It is memory of the sequence, not of arithmetic.

F. Albanese energies and exact checks

Proposition 1 holds on all 82 graphs: the backbone winding has energy exactly N, and on the coarse graphs $\kappa(Y^*)/(1+R)$ is an integer in 82 of 82 cases (PARI/GP, exact arithmetic). The closure flux reduces to the end-to-end resistance R (Fig. 3). In the relaxed globules $R = 0.32 \pm 0.17$ (true) and 0.30 ± 0.07 (S1), indistinguishable. In the partially collapsed production globules R spans more than two decades and its median is lower for the true chain than for S1, but no contrast survives correction (smallest $p_{\mathrm{Holm}} = 0.16$, S6″ versus S1). The effective mass of the closed network is set by how well its two ends are connected, which is a property of the collapse, not of the sequence.

V. DISCUSSION

The constructions of algorithmic class field theory transfer to the contact network of a folded polymer and are exactly computable on it, but the arithmetic content that makes them interesting on arithmetic quotients does not transfer with them. On the Bruhat–Tits tree the sandpile order is a product of Frobenius traces and the bulk channel carries automorphic data [5]. On a contact network the sandpile group is a large, generic abelian group whose structure does not depend on the sequence that drove the collapse. What the bulk channel registers is the sequence *as a function on the vertices*, through the energy of a divisor. That energy is dominated by the chain-scale modes, which for the primes are the PNT envelope.

The two-channel picture nevertheless survives in a reduced form. The bulk channel separates the envelope-carrying arms from the shuffle; the boundary channel separates nothing. This is the dichotomy of Ref. [5] with its arithmetic content replaced by positional content, and it matches the conclusion of the companion paper: polymer readouts see the arithmetic of a prime sequence through its envelope and its composition, and through nothing finer [3]. The eight-globule pilot that suggested otherwise did not replicate, and we report it as a cautionary example of how a random-effects design and a family-wise correction change the reading of a marginal p-value.

What every channel does see is memory. A collapsed globule is adapted to its own stiffness sequence: its con-

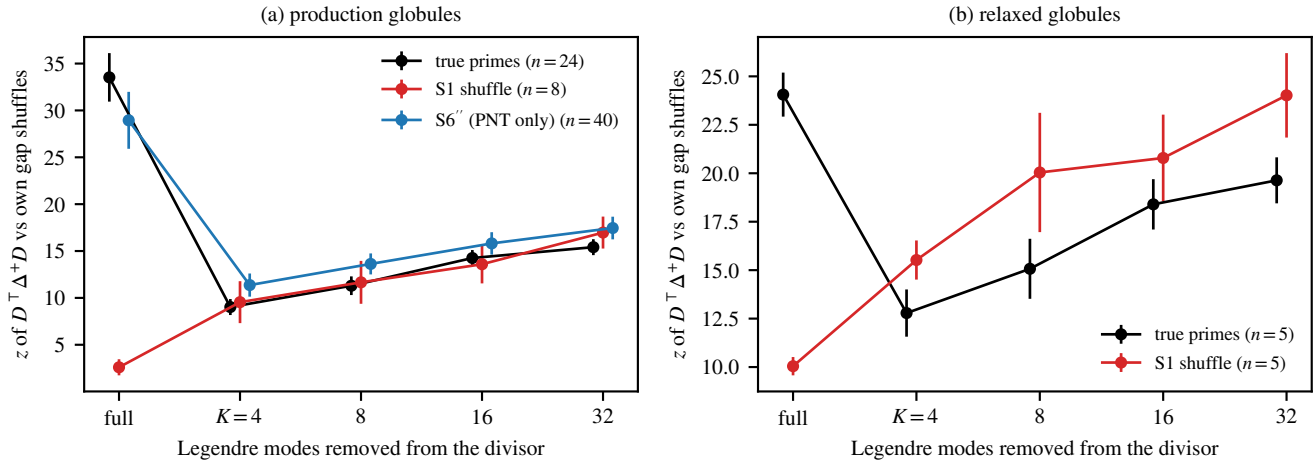


FIG. 1. Bulk channel. z of the divisor energy against each globule's own gap shuffles (mean ± s.e.m.), for the full divisor and with Legendre modes up to degree K removed. (a) Production globules; (b) relaxed globules. The full divisor separates the envelope-carrying arms (true, S6″) from S1; removing the smooth modes removes every arm difference.

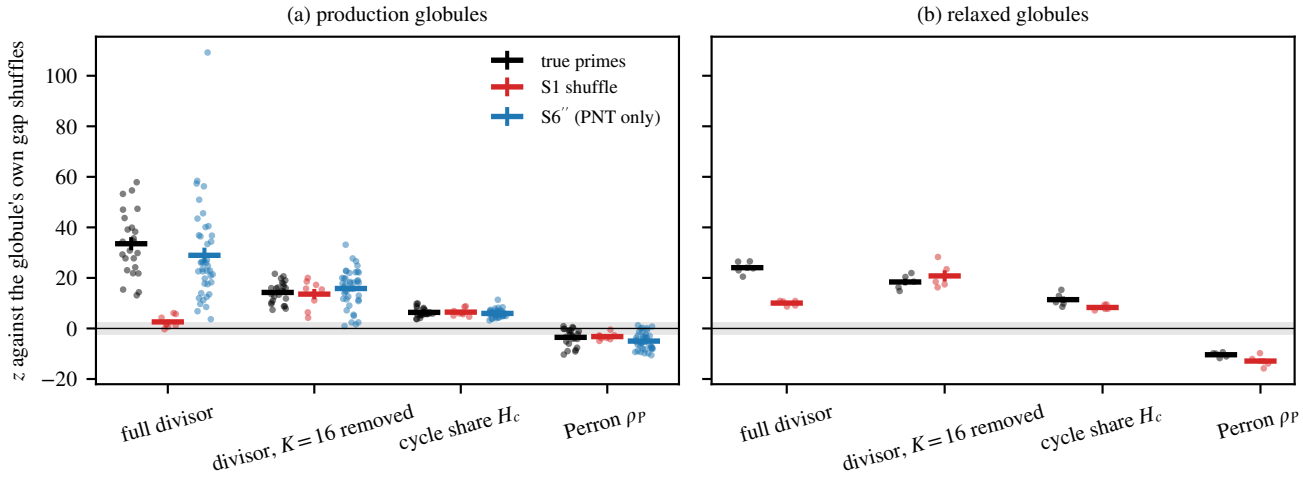


FIG. 2. Own-sequence memory. Per-globule z against 20 gap shuffles on the same conformation, for the full divisor energy, the envelope-removed divisor energy ($K = 16$), the contact cycle share H_c and the Perron correlation ρ_P (dots: globules; bars: mean ± s.e.m.; grey: $|z| < 2$). (a) Production; (b) relaxed. Panel (b) has no S6″ arm because the 10^6-step isothermal hold was run only for the true chain and the S1 shuffle [2].

tacts, cycles and Perron mode are all far from what the same conformation would give a permuted sequence. This is a property of folding, not of the primes, and it is equally strong for shuffles. It may be useful elsewhere. For contact maps measured in experiment, the own-sequence z of a divisor energy is a parameter-free measure of how strongly a conformation is adapted to a sequence-dependent property of its monomers.

The series in one picture. This paper closes a four-part study of what a polymer folded under a prime-programmed stiffness can register of the primes [1–3]. The picture it leaves is simple. The fold has a strong topological memory of its own sequence: its radial organization, its contacts, its cycles and its Perron mode are all adapted to the stiffness sequence that drove the collapse. That memory is generic, as strong for a random permutation of the gaps as for the primes. Its response to number-theoretic detail is confined to two channels: the smooth prime-number-theorem envelope, read through the position of each bead along the chain, and the sieve composition of the gap multiset, which converges with sieve depth. Correlation lengths that grow with the chain, congruence confinement, and every finer arithmetic structure we tested lie outside what the fold reads.

Limitations. The contact graph depends on the cutoff

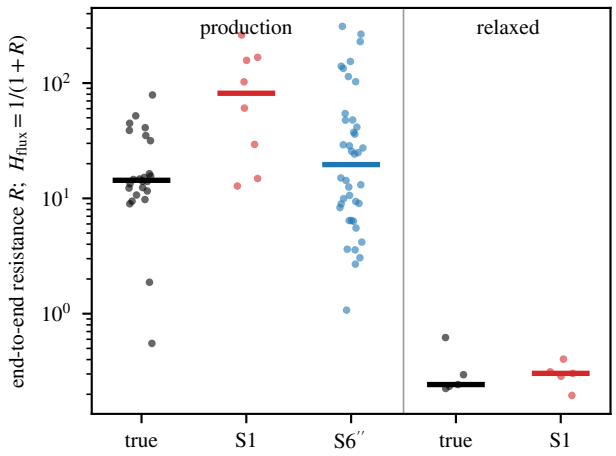


FIG. 3. End-to-end effective resistance R of the contact network (log scale; bars: medians). The closure-flux Albanese energy is $1/(1+R)$ (Proposition 1).

r_c, which we did not vary. The production globules are partially collapsed coils, and only ten relaxed globules exist. The S1 arm has eight production globules. Exact sandpile arithmetic is computed only on coarse-grained graphs. The S6″ surrogate matches the leading Legendre coefficients of the true composition to 0.93–1.0 [1], so residual envelope differences at the few-percent level cannot be excluded at $K \leq 4$; at $K \geq 8$ they are removed by construction.

VI. CONCLUSION

On 82 conformations of a prime-programmed polymer, the invariants of the contact network fall into three groups. Unweighted invariants, including the exact sandpile group, are blind to the sequence. Sequence-weighted bulk invariants see the prime-number-theorem envelope, and boundary invariants see no arm difference: the two-channel dichotomy of algorithmic class field theory, with positional in place of arithmetic content. Every channel records a strong memory of the globule's own sequence that is identical for primes, shuffles and PNT-only surrogates. Nothing in the contact network distinguishes the primes from a smooth model of their density once the envelope is removed; a marginal pilot signal to the contrary did not replicate.

ACKNOWLEDGMENTS

Code, data, tests and figure scripts are available at https://github.com/Ruqing1963/contact-network-two-channels and archived at doi:10.5281/zenodo.23100245. The conformations are those of the PDTP repository, version 4.0 [1].

[1] R. Chen, Beyond the marginal distribution: prime-gap ordering imprints on a self-avoiding polymer through the small-prime end of the chain (2026), version 4.0, Zenodo, doi:10.5281/zenodo.22944122.
[2] R. Chen, Collapse, convergence, and the scale of the prime-gap imprint on polymer dynamics (2026), Zenodo, doi:10.5281/zenodo.23076433.
[3] R. Chen, What a prime-programmed polymer can read: sieve composition, congruence traps, and arithmetic blindness (2026), Zenodo, doi:10.5281/zenodo.23094095.
[4] E. Lieberman-Aiden *et al.*, Comprehensive mapping of long-range interactions reveals folding principles of the human genome, Science **326**, 289 (2009).
[5] R. Chen, Algorithmic non-commutative class field theory: Hecke operators, sandpile groups and operator algebras on trees and buildings, with applications (2026), Zenodo, doi:10.5281/zenodo.22883908.
[6] N. L. Biggs, Chip-firing and the critical group of a graph, J. Algebraic Combin. **9**, 25 (1999).
[7] L.-H. Lim, Hodge Laplacians on graphs, SIAM Rev. **62**, 685 (2020).
[8] D. J. Klein and M. Randić, Resistance distance, J. Math. Chem. **12**, 81 (1993).
[9] S. Holm, A simple sequentially rejective multiple test procedure, Scand. J. Stat. **6**, 65 (1979).
[10] Y. Benjamini and Y. Hochberg, Controlling the false discovery rate: a practical and powerful approach to multiple testing, J. R. Stat. Soc. B **57**, 289 (1995).